amazon rme

AMAZON BASE BUILDING MAINTENANCE STANDARD

GLOBAL CENTRAL RME STANDARDS & COMPLIANCE

Task Title	ELECTRICAL SYSTEM, HIGH VOLTAGE, TRANSFORMERS - 11
APM Object	ELEC.HV.TX
APM EC	AMZ.ELEC.HV.TX.GRP
BBM Handbook Title	High Voltage Transformers
General tasks always required	Follow safety procedures - e.g. 1MRA, LOTO, PPE, etc. Ensure that all WOs are carried out by qualified, certified staff with appropriate familiarisation and supervision as required Add notes for each observation and create a FWO if necessary to eliminate defects Use appropriate tools, chemicals and industry standard practices (RME policy permitting) Ensure relevant logs, readings and certification is uploaded to APM prior to WO closure These tasks are based on industry guidance, SME knowledge and HIE learning. Please ensure vendor specific requirements are followed. To suggest improvements please raise a SIM SIM - Link to SIM template

Last Update: (Version)	Jun-24	**Geo Applied:**	Intl APM Instance	**F-Gas**	Not Identified	**Activies may be covered by No-Go restrictions:**	Yes	**Special Certificaion Required:**	TBC	**Delivered By:**	Carried out by trained technician certified in the MV electrical field	**OCB:**	BBM
Evidence for CT:	NO	**Pricing template Position:**	11	**Related CT Areas:**	CT.ELEC.TROILTEST CT.ELEC.ELECINST	**Assets included in this PM:**	OIL TRANSFORMER - DRY TRANSFORMER	**Schedule With:**	ELEC.HV.SWGR				

Description of System / Technology (According to BBM Handbook)

Transformers are used in electric circuits to change the voltage of electricity flowing in the circuit. It is used to increase the voltage (called 'stepping up') or decrease the voltage ('stepping down'). There are 2 types of transformer oil filled or dry-type that need differentiate by nameplate.

Additional Information

All sites with the above listed Base Building Asset should conduct routine maintenance and inspections according to this standard PM package. The Amazon maintenance standard supports compliance, reduces safety risks,improves asset reliability & efficiency. Some maintenances captures several assets within one PM package -> Ensure that corrective/reactive maintenance for assets where proactive maintenance is contained under another asset is created against the original asset.

Task Index	Component	Tasks	Tasks Nested for frequencies > weekly								
			Daily	Weekly	Monthly	3 Monthly	6 Monthly	12 Monthly	24 Monthly	36 Monthly	48 Monthly	60 Monthly
1	Station outside check	Is the entrance freely accessible & free of obstacles			SC			PM				
2	Station outside check	Are there any visual damage or excessive vegetation at the building or around			SC			PM				
3	Station outside check	Check roof & roof drainage - clean drain inlets if necessary			SC			PM				
4	Station outside check	Check if the lightning protection is fixed appropriately & not damaged			SC			PM				
5	Station outside check	Are there any damage or corrosion at the cable trays			SC			PM				
6	Station outside check	Is the door locked properly - protection against unauthorized entry			SC			PM				
7	Station inside check	Does the light work properly			SC			PM				
8	Station inside check	Is the heater working			SC			PM				
9	Station inside check	Check heater for dirt, damage, corrosion			SC			PM				
10	Station inside check	Are there any water leaks visible at the walls or ceiling			SC			PM				
11	Station inside check	Are there any damage at the walls, ceiling or transformer cage			SC			PM				
12	Station inside check	Is the transformer cage door locked properly - protection against unauthorized entry			SC			PM				
13	Station inside check	Are technical installations damaged or loose (e.g. cable trays, MCP, fire extinguisher)			SC			PM				
14	Station inside check	Are flammable or unnecessary goods stored in the room			SC			PM				
15	Station inside check	Check the function of the room ventilation - natural & mechanical			SC			PM				
16	Safety	Are all relevant safety notices visible and free from damage			SC			PM				
17	Safety	Is appropriate fire-fighting equipment available in the rooms			SC			PM				
18	Safety	Is a rescue pole available in the room and in good condition?			SC			PM				
19	Safety	Is insulated floor matting installed and in good condition?			SC			PM				
20	Safety	Is HV detection pole available and in good working order?			SC			PM				
21	Safety	Inspect and test all tools, test instruments to ensure serviceable condition						PM				
22	Safety	Are disconnecting devices lockable in the open position (LOTO)						PM				
23	Documentation	Are the circuit diagrams and relevant technical documentation available and complete						PM				
24	Documentation	Are technical drawings hanging at the wall						PM				
25	Thermographic scan	Perform thermographic scans of the whole system (e.g. Busbars & Transformer, etc.) -> Create a FWO for over-temperature to determine the root cause immediately						PM				
26	Thermographic scan value	Add a detailed note for all over-temperature measurements -> list the location and temperature-> Upload the thermographic scan						PM				
27	Shut down	De-energize the transformer						PM				
29	Shut down	Measure whether the system is voltage-free						PM				
30	Shut down	Perform earthing & short-circuit of all phases against each other & Protection Earth (ground)						PM				
31	Shut down	Block of neighboring live parts						PM				
32	Transformer	Visual check of the transformer for dirt, damage, corrosion & oil leakage, etc.						PM				
33	Transformer	Carry out functional cleaning						PM				
34	Transformer	Check breathers for correct function & damage (dehumidifier)						PM				
35	Transformer	Check transformer protection for correct function and damage (e.g. Buchholz relay)						PM				
36	Transformer	Inspect all cable and busbar terminations for signs of overheating						PM				

37	Transformer	Ensure all cable & busbar connections are correctly torqued according to manufacturers specification						PM					
38	Transformer	Inspect all cables and connections insulation/shields for damage						PM					
39	Transformer	Check if the system is properly connected to the PE (ground), check also for damage & corrosion						PM					
40	Transformer	Are all devices properly labeled or marked						PM					
41	Final Check	Check the O&M manual and carry out additional required tasks						PM					
42	Re-energize	Remove barriers of neighboring live parts						PM					
43	Re-energize	Remove earthing & short-circuit the equipment						PM					
45	Re-energize	Energize the transformer						PM					
46	Re-energize	Measure system voltage using appropriate tools						PM					

Change Log

June 2024 - New PM according to the last OCB changes. Removed task regarding LOTO.